

Phase 6 TSP Replay Mechanism Protocol

This document defines the conservative follow-up after the completed phase-5 routing study.

The scientific motivation is narrow and mechanistic:

On symmetric TSPLIB95 TSP, why did random_replay beat failure_replay?

The purpose of phase 6 is not to claim a stronger result than phase 5. The purpose is to explain the surprising phase-5 result with a cleaner TSP-only experiment.

Central Question

The main phase-6 question is:

Is replay helping because it remembers true failures, or because it preserves broad distributional coverage?

The strongest phase-6 result shape is:

- replay conditions with higher archive descriptor diversity also show better held-out transfer
- raw failure severity alone is a weaker predictor than archive diversity
- residual-failure replay improves on naive failure replay by separating "hard for everyone" from "specifically bad for this heuristic"

Descriptor Basis

Phase 6 adds per-instance descriptors for all train, holdout, replay, and adversarial TSP instances:

- number of cities
- coordinate spread
- nearest-neighbor distance statistics
- MST length
- convex-hull ratio
- clustering and cluster-separation scores
- grid-like versus random-like structure
- bottleneck or two-cluster structure
- nearest-neighbor trap score

Research basis:

- Reinelt, TSPLIB-A Traveling Salesman Problem Library (<https://doi.org/10.1287/ijoc.3.4.376>), 1991
- Smith-Miles and Lopes, Measuring instance difficulty for combinatorial optimization problems (https://www.cs.ubc.ca/labs/algorithms/EARG/stack/2012_ComputersAndOperationResearch_SmithMiles_MeasuringInstanceDifficulty.pdf), 2012
- Kerschke et al., A study on the effects of normalized TSP features for automated algorithm selection (<https://www.sciencedirect.com/science/article/pii/S0304397522006089>), 2022
- Rice's algorithm-selection framing, as summarized in Smith-Miles and Lopes above, motivates using instance descriptors plus baseline performance to estimate expected difficulty

- Replication discipline in this repository still follows the repeated-run guidance already used in earlier phases:
- Deep Reinforcement Learning That Matters (<https://arxiv.org/abs/1709.06560>)
- Reliable: Better Evaluation for Reinforcement Learning (<https://arxiv.org/abs/2108.13264>)

Benchmark Basis

Phase 6 stays TSP-only and keeps the same benchmark basis as phase 5 for comparability:

- the official TSPLIB95 Euclidean train subset
- held-out TSPLIB95 Euclidean instances
- adversarial synthetic geometric layouts
- exact synthetic geometric holdouts

This is deliberate. The phase-6 question is about replay mechanism, not cross-problem transfer.

Replay Arms

The official comparison set is:

1. no_replay
2. random_replay
3. failure_replay
4. random_replay_compression
5. stratified_random_replay
6. diversity_weighted_replay
7. residual_failure_replay
8. diversity_failure_replay
9. diversity_failure_replay_compression

Key additions relative to phase 5:

- random_replay_compression, because the winning replay arm from phase 5 also needs a compression test
- residual_failure_replay, because raw failure gap is confounded with baseline hardness
- descriptor-aware replay variants, because archive coverage itself may be the relevant mechanism

Residual Failure Replay

The defining phase-6 mechanism is:

$\text{residual_failure_gap} = \text{observed_gap} - \text{expected_gap}(\text{instance_features}, \text{baseline_portfolio})$

The expected gap is estimated from:

- the instance descriptors above
- a fixed baseline heuristic portfolio
- descriptor-nearest neighbors over the benchmark and adversarial reference instances

Interpretation:

- large raw gap on a hard instance is not automatically a replay-worthy "true failure"
- large positive residual gap is stronger evidence that the current evolved heuristic has a specific weakness

Required Logging

Log the replay archive composition over time, including:

- archive descriptor diversity
- archive hardness
- archive residual hardness
- archive size bias
- replay failure concentration
- structure histogram
- size histogram

These should be stored per epoch and summarized in both condition-level and aggregate reports.

Endpoints

Primary endpoint:

- final held-out TSPLIB mean optimality gap

Secondary endpoints:

- final synthetic transfer gap
- combined transfer gap
- archive descriptor diversity
- archive hardness
- archive size bias
- replay failure concentration
- accepted code novelty
- heuristic complexity
- adaptation efficiency

Key diagnostic plot:

- held-out gap versus replay-archive descriptor diversity

The target interpretation is whether descriptor diversity predicts transfer better than raw archive hardness.

Interpretation Rules

- Do not claim that residual replay is better unless it beats raw failure replay on held-out transfer, not only training.
- Do not claim that archive hardness is the right explanation if diversity tracks transfer better.

- Do not claim operator discovery or algorithm discovery from phase 6; this phase still evolves whole solver scaffolds.
- If random replay remains best, treat that as evidence about distributional coverage and anti-thrashing, not as a failure of the whole replay hypothesis.

Replication

Official main-study target:

- 20 paired seed offsets with the fixed list 0, 1000, 2000, ..., 19000

Use paired comparisons across those same offsets and report uncertainty on paired deltas, not only per-condition means.

Deliverable

The required deliverable is a short mechanism report that answers:

1. Why did random_replay beat failure_replay?
2. Does replay-archive diversity explain transfer better than replay hardness?
3. Does compression help once applied to the best replay mechanism?
4. Are the final heuristics genuinely different, or just differently tuned?

Operational Entry Points

- Prepare the benchmark manifest with `prepare_tsp_benchmarks.py`
- Run the suite with `run_tsp_suite.py` and `configs/tsp_phase6_suite/01_replay_mechanism.json`
- Use `configs/tsp_phase6_suite/RUNBOOK.md` for concrete commands
- Aggregate repeated runs with `aggregate_tsp_phase6_runs.py`
- Official results note: versioned on branch `replay-mechanism` as `docs/PHASE_6_TSP_REPLAY_MECHANISM_RESULTS_2026-05-20.md`

Relationship To Phase 5

Phase 5 established that the first routing pass was mixed. Phase 6 is the mechanism follow-up that explains that result before the project makes stronger claims about replay-aware abstraction.